

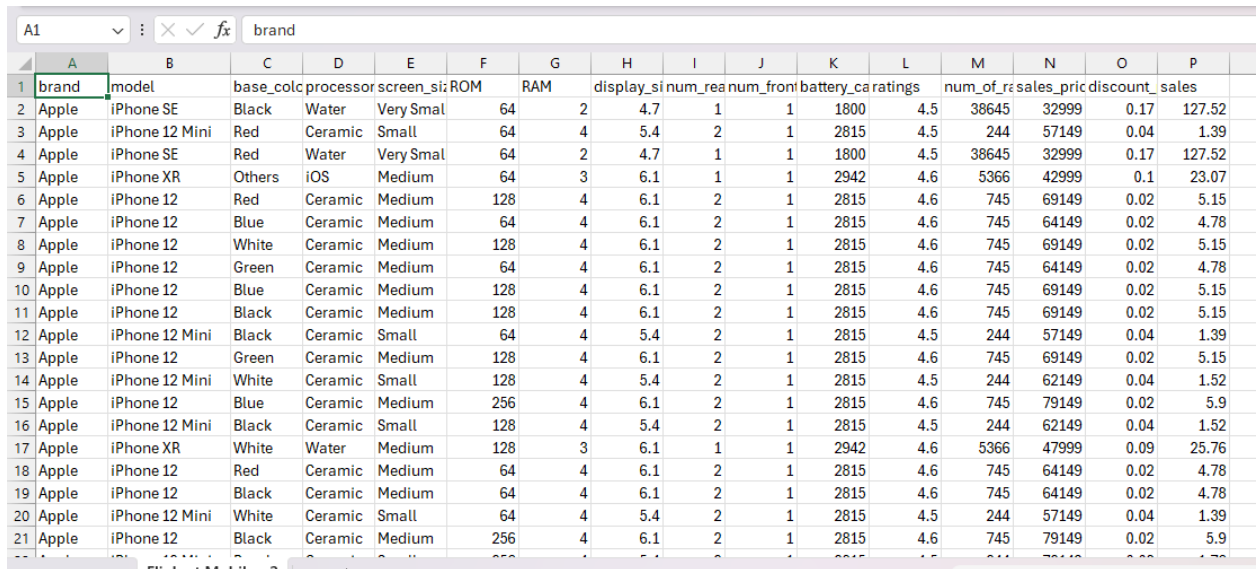
“Flipkart Mobile Sales Analysis”
Business Information Visualization & Analysis Case Study

By: Amishi Desai, Chahel Gupta

Part A: Data Selection

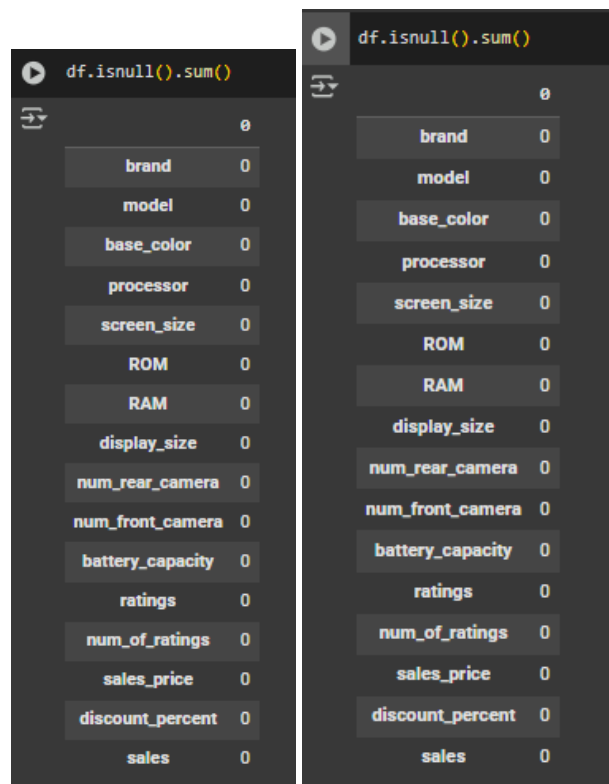
For this project of ours, we have used Flipkart Mobile Sales dataset. This dataset includes data on mobile phones of some of India's topmost mobile brands: Apple, Poco, Samsung, Realme and Xiaomi. This dataset comprises attributes such as: Brand, Model Name, Base Color, Processor, Screen Size, ROM, RAM, Display Size, number of rear camera, number of front camera, battery capacity, number of ratings, sales, price, discounts and the sales. This dataset has at least one attribute which distinguishes each product and it has no null values.

Source: <https://www.kaggle.com/datasets/shubhambathwal/flipkart-mobile-dataset>



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	brand	model	base_color	processor	screen_size	ROM	RAM	display_size	num_rear_camera	num_front_camera	battery_capacity	ratings	num_of_ratings	sales_price	discount_percent	sales
2	Apple	iPhone SE	Black	Water	Very Small	64	2	4.7	1	1	1800	4.5	38645	32999	0.17	127.52
3	Apple	iPhone 12 Mini	Red	Ceramic	Small	64	4	5.4	2	1	2815	4.5	244	57149	0.04	1.39
4	Apple	iPhone SE	Red	Water	Very Small	64	2	4.7	1	1	1800	4.5	38645	32999	0.17	127.52
5	Apple	iPhone XR	Others	iOS	Medium	64	3	6.1	1	1	2942	4.6	5366	42999	0.1	23.07
6	Apple	iPhone 12	Red	Ceramic	Medium	128	4	6.1	2	1	2815	4.6	745	69149	0.02	5.15
7	Apple	iPhone 12	Blue	Ceramic	Medium	64	4	6.1	2	1	2815	4.6	745	64149	0.02	4.78
8	Apple	iPhone 12	White	Ceramic	Medium	128	4	6.1	2	1	2815	4.6	745	69149	0.02	5.15
9	Apple	iPhone 12	Green	Ceramic	Medium	64	4	6.1	2	1	2815	4.6	745	64149	0.02	4.78
10	Apple	iPhone 12	Blue	Ceramic	Medium	128	4	6.1	2	1	2815	4.6	745	69149	0.02	5.15
11	Apple	iPhone 12	Black	Ceramic	Medium	128	4	6.1	2	1	2815	4.6	745	69149	0.02	5.15
12	Apple	iPhone 12 Mini	Black	Ceramic	Small	64	4	5.4	2	1	2815	4.5	244	57149	0.04	1.39
13	Apple	iPhone 12	Green	Ceramic	Medium	128	4	6.1	2	1	2815	4.6	745	69149	0.02	5.15
14	Apple	iPhone 12 Mini	White	Ceramic	Small	128	4	5.4	2	1	2815	4.5	244	62149	0.04	1.52
15	Apple	iPhone 12	Blue	Ceramic	Medium	256	4	6.1	2	1	2815	4.6	745	79149	0.02	5.9
16	Apple	iPhone 12 Mini	Black	Ceramic	Small	128	4	5.4	2	1	2815	4.5	244	62149	0.04	1.52
17	Apple	iPhone XR	White	Water	Medium	128	3	6.1	1	1	2942	4.6	5366	47999	0.09	25.76
18	Apple	iPhone 12	Red	Ceramic	Medium	64	4	6.1	2	1	2815	4.6	745	64149	0.02	4.78
19	Apple	iPhone 12	Black	Ceramic	Medium	64	4	6.1	2	1	2815	4.6	745	64149	0.02	4.78
20	Apple	iPhone 12 Mini	White	Ceramic	Small	64	4	5.4	2	1	2815	4.5	244	57149	0.04	1.39
21	Apple	iPhone 12	Black	Ceramic	Medium	256	4	6.1	2	1	2815	4.6	745	79149	0.02	5.9

Figure 1: Flipkart Mobile Sales Dataset



df.isnull().sum()	0
brand	0
model	0
base_color	0
processor	0
screen_size	0
ROM	0
RAM	0
display_size	0
num_rear_camera	0
num_front_camera	0
battery_capacity	0
ratings	0
num_of_ratings	0
sales_price	0
discount_percent	0
sales	0

df.isnull().sum()	0
brand	0
model	0
base_color	0
processor	0
screen_size	0
ROM	0
RAM	0
display_size	0
num_rear_camera	0
num_front_camera	0
battery_capacity	0
ratings	0
num_of_ratings	0
sales_price	0
discount_percent	0
sales	0

Figure 2: Dataset having zero null values

There are 16 columns and 430 rows, wherein each column has a self-explanatory title and each row has a mobile with at least one distinct feature.

Description of the attributes:

Attribute	Description	Data Type
brand	Name of the brand	Categorical
model	Name of the model	Categorical
base_color	Color of the phone	Categorical
processor	The processor brand used	Categorical
screen_size	Categorical screen size	Categorical
ROM	ROM in gigabyte	Numeric – Discrete
RAM	RAM in gigabyte	Numeric – Discrete
display_size	Size of display in inches	Numeric – Continuous
num_rear_camera	Number of cameras on back	Numeric – Discrete
num_front_camera	Number of cameras on front	Numeric – Discrete
battery_size	Battery in mAH	Numeric – Continuous
ratings	Rating by the customer	Numeric – Continuous
num_of_ratings	Number of people who have rated the product	Numeric – Continuous
sales_price	Selling price after discount	Numeric – Continuous
discount_percent	Discount offered in percentage	Numeric – Continuous
sales	Sales of the product in crore product	Numeric – Continuous

Initial Questions:

- Q1. Which mobile brands dominate the Flipkart platform in terms of listings and ratings?
- Q2. Do technical specifications like battery or the camera quality influence the price?
- Q3. Is there a relationship between price and average rating?
- Q4. Is there any brand that offers high specs at below average prices?
- Q5. Do technical specifications like battery or camera quality significantly influence the price?

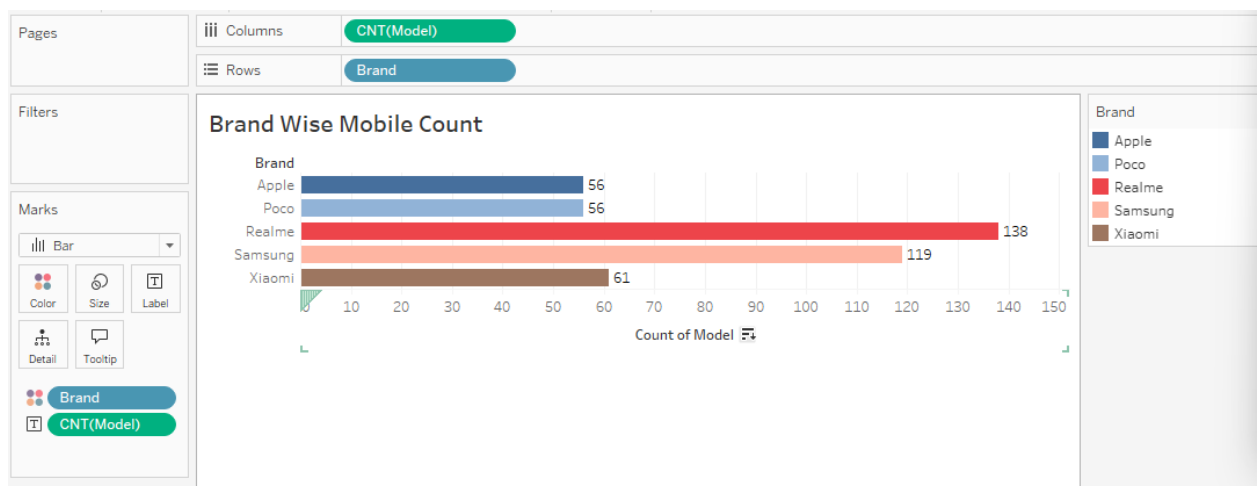
Part B: Exploratory Visual Analysis

Phase 1: Observations

- **No Null Values:** All columns have 430 non-null entries.
- **Data Types:**
 - Categorical values (brand, model, processor, base_color, screen_size) are of object type.
 - Continuous/numerical variables are given by int64 or float64.
- **Relationships to be explored:**
 - We might explore if higher RAM/ROM correlates with price, ratings, or sales.
 - Check if discounts significantly impact sales.
 - Are there standout brands/models with unusual performance?
- **Expected Patterns to Validate:**
 - Higher battery capacity → Higher ratings?
 - Larger display → Higher price?
 - Discounts → Boosted sales?

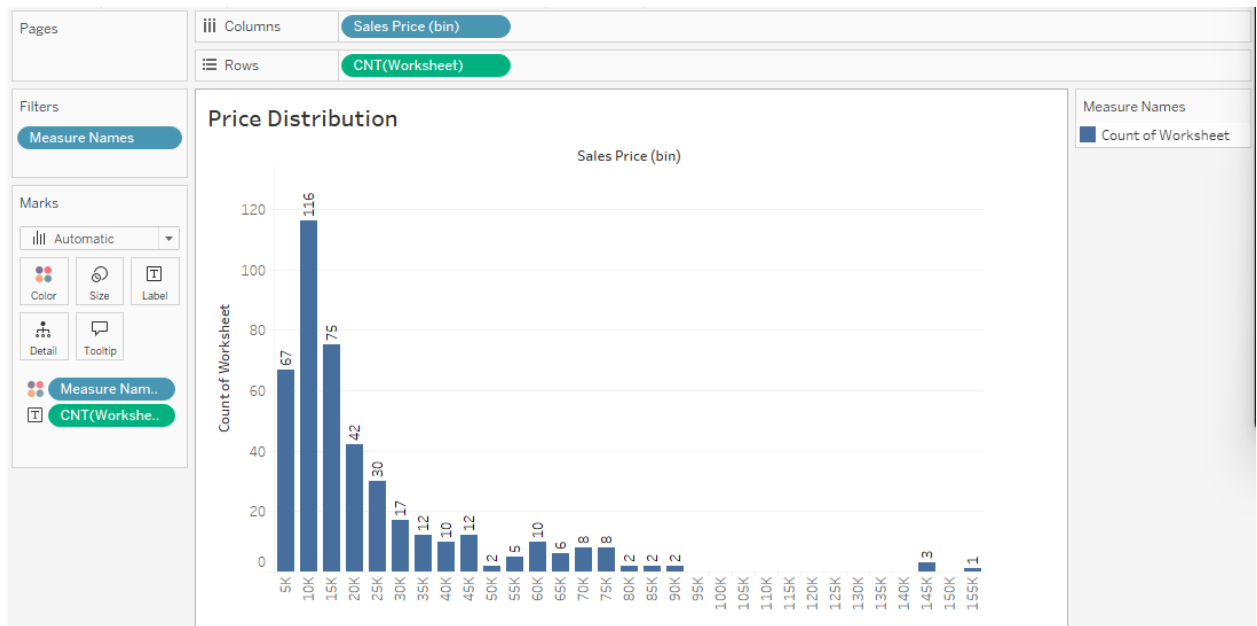
Phase 2: Visualization through charts

Q. Which mobile brand has the highest number of listings on Flipkart?



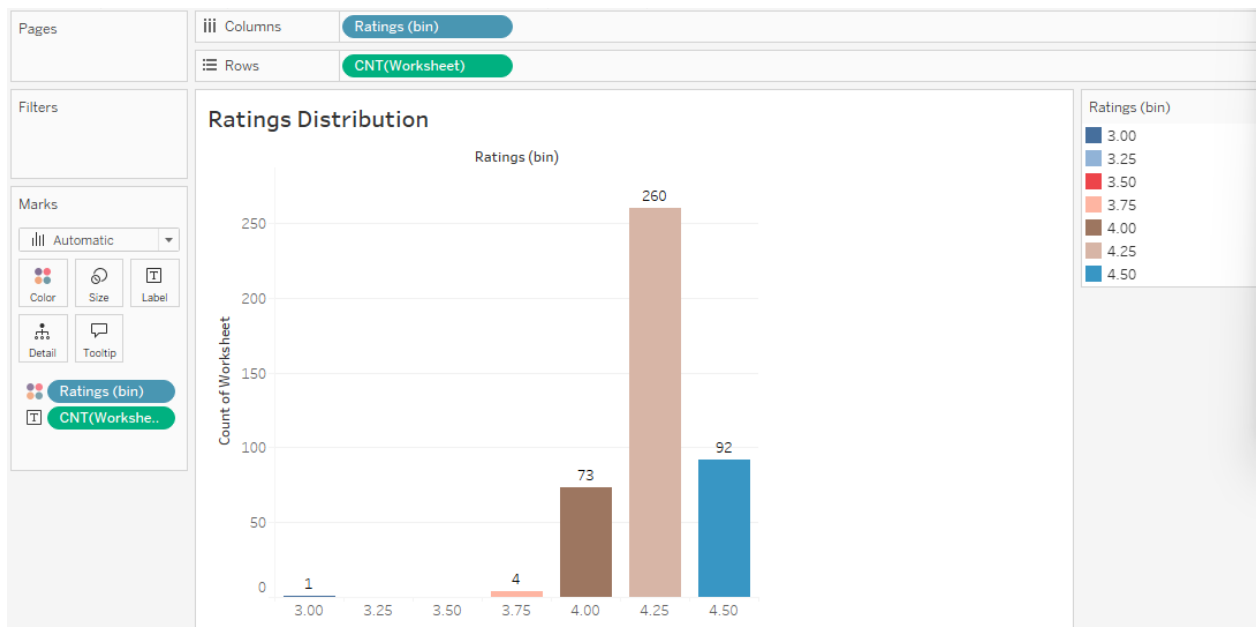
Insights: The graph shows that Realme has the most listings, followed by Samsung indicating its strong presence and product variety on the platform.

Q. How are mobile phones distributed across different price ranges?



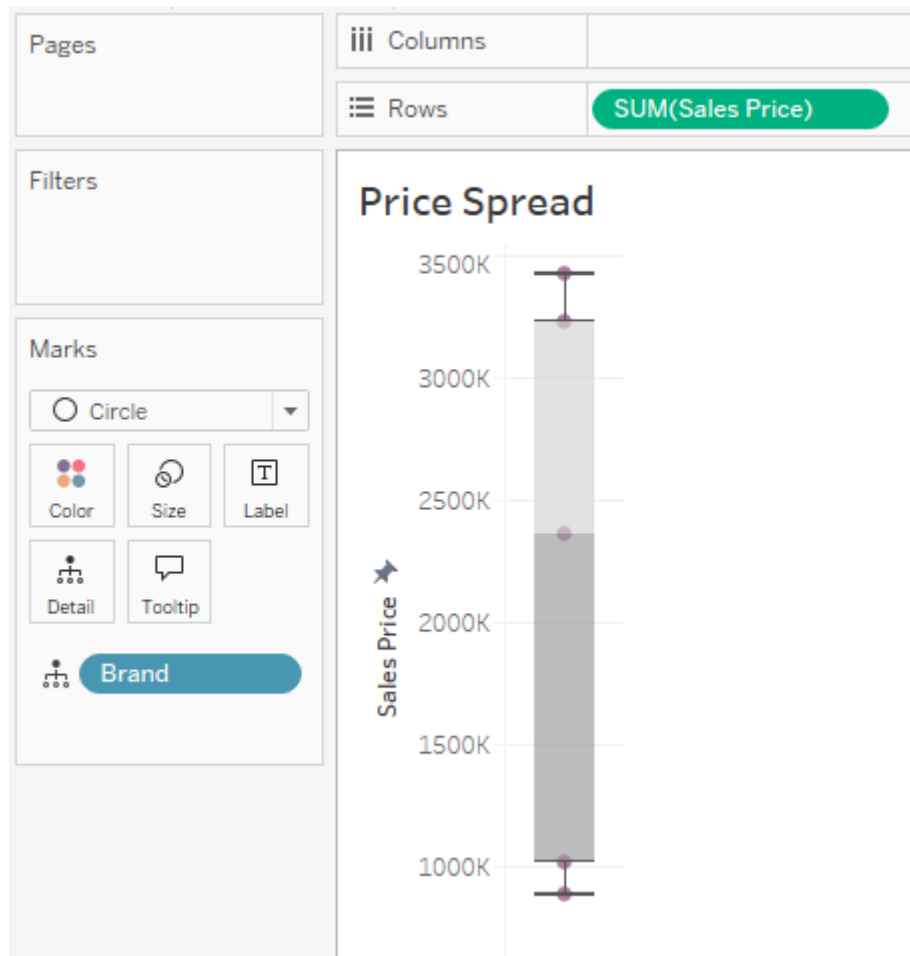
Insights: The bar graph shows that the price distribution of the various mobile phones listed on flipkart are generally skewed towards the lower end, suggesting a popularity of affordable phones amongst its user.

Q. What is the distribution of customer ratings among listed mobile phones?



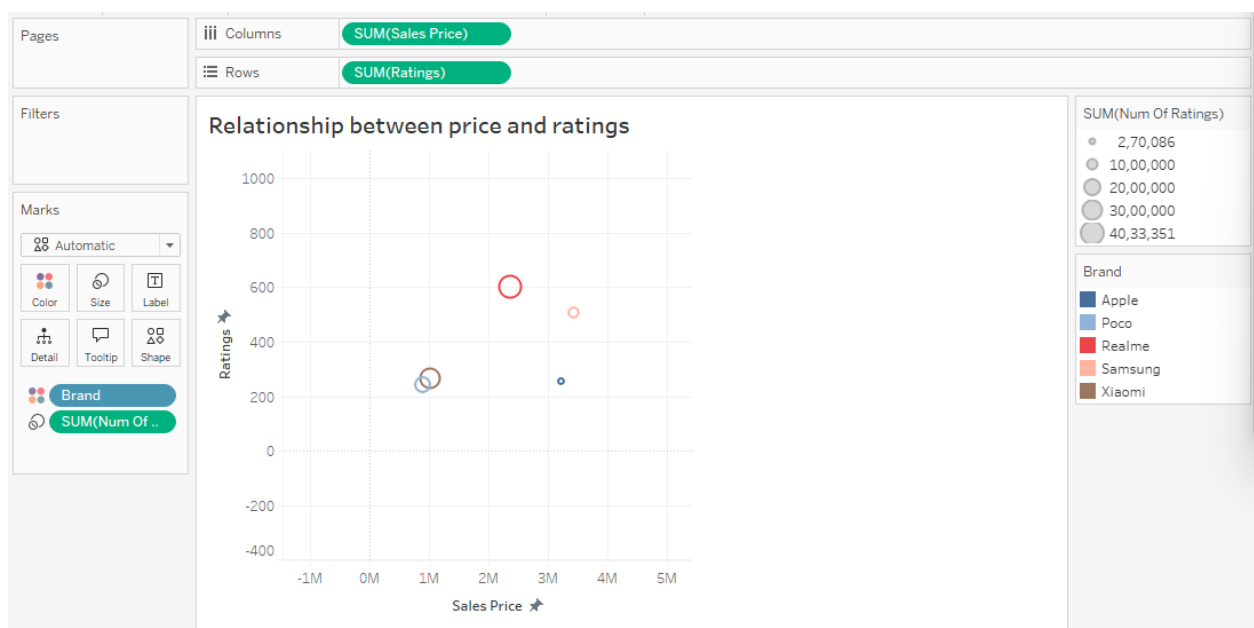
Insights: The rating distribution suggests that most of flipkart's customers are satisfied with their purchases as the number of ratings 4.0 to 4.5 are high.

Q. What is the range, median, and outliers in mobile phone prices?



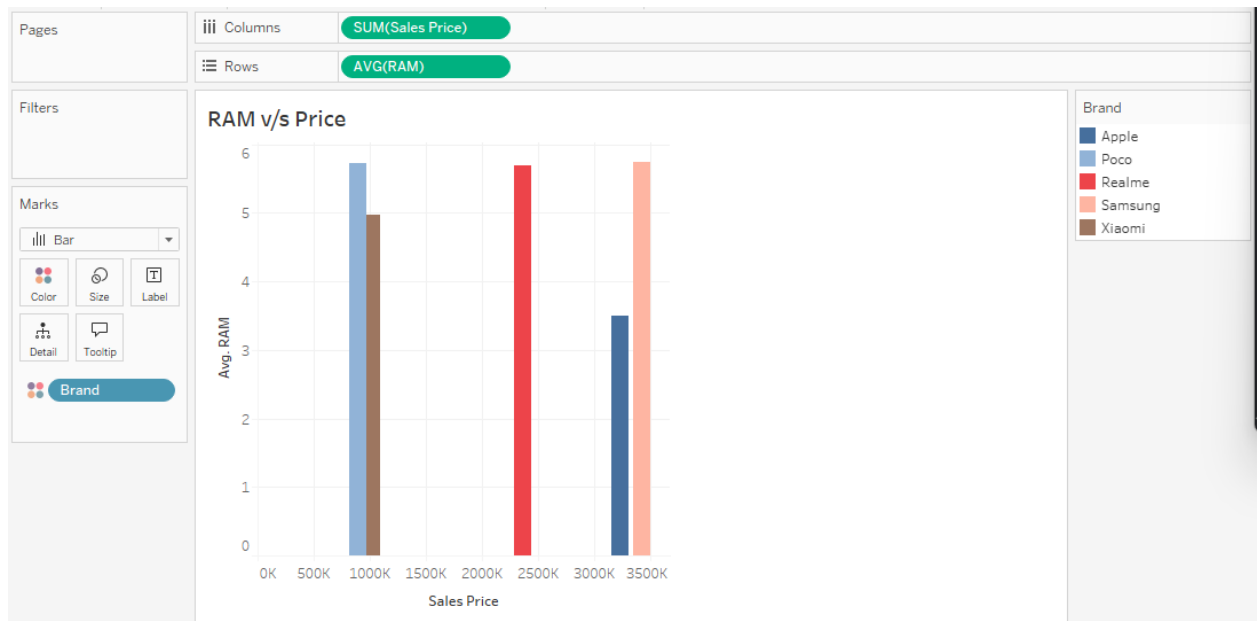
Insights: The box plot reveals a wide spread in mobile prices, ranging from ₹90K to ₹350K. Most phones are priced between ₹100K and ₹300K, with a median around ₹230K. This suggests a diverse pricing strategy targeting both budget and premium segments.

Q. Is there any correlation between the price of a phone and its customer rating?



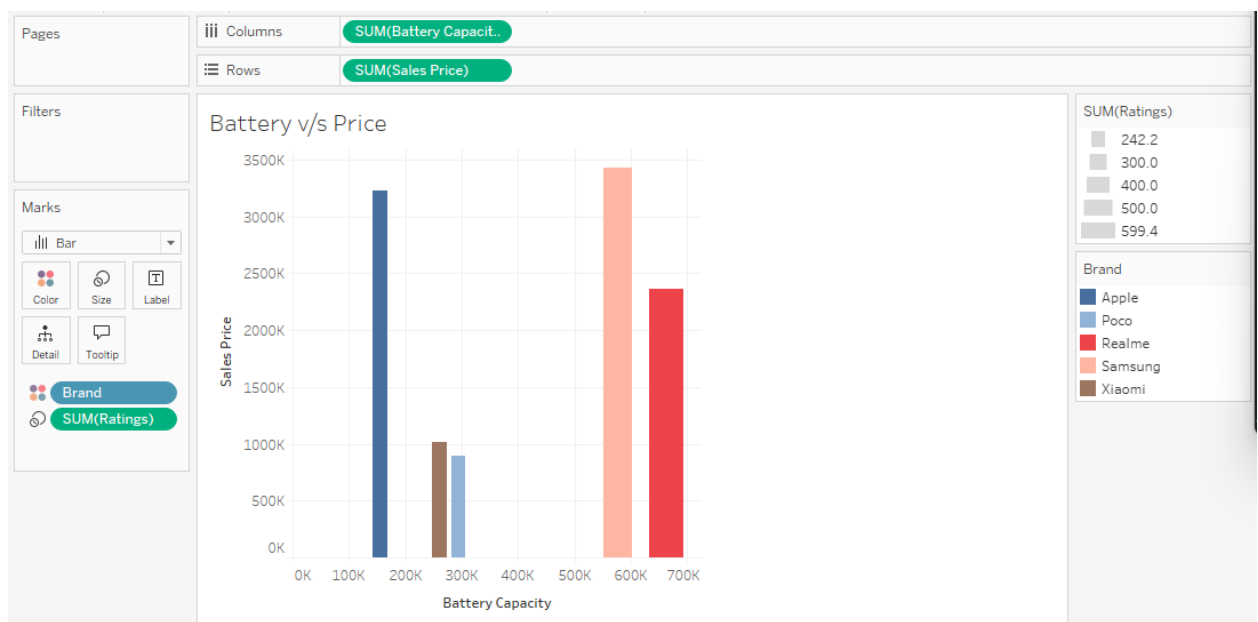
Insights: There is a weak positive correlation between price and customer ratings, implying that higher price doesn't always guarantee better ratings.

Q. How does the amount of RAM in a phone impact its pricing?



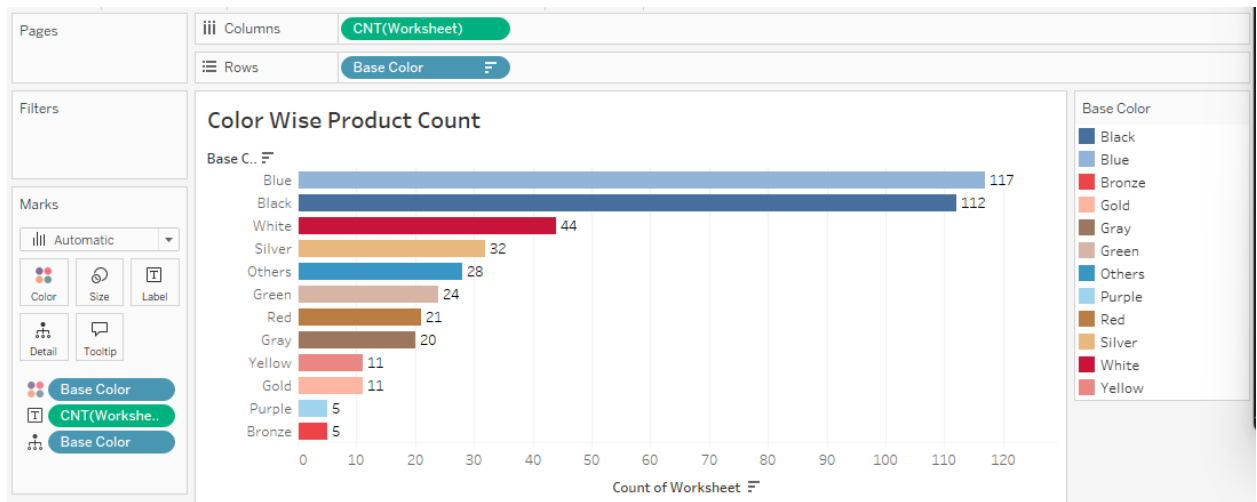
Insights: As expected, phones with higher RAM (especially 8GB and above) tend to be priced higher, showing a direct relationship between performance specs and cost.

Q. What is the relationship between battery capacity and the price of a mobile phone?



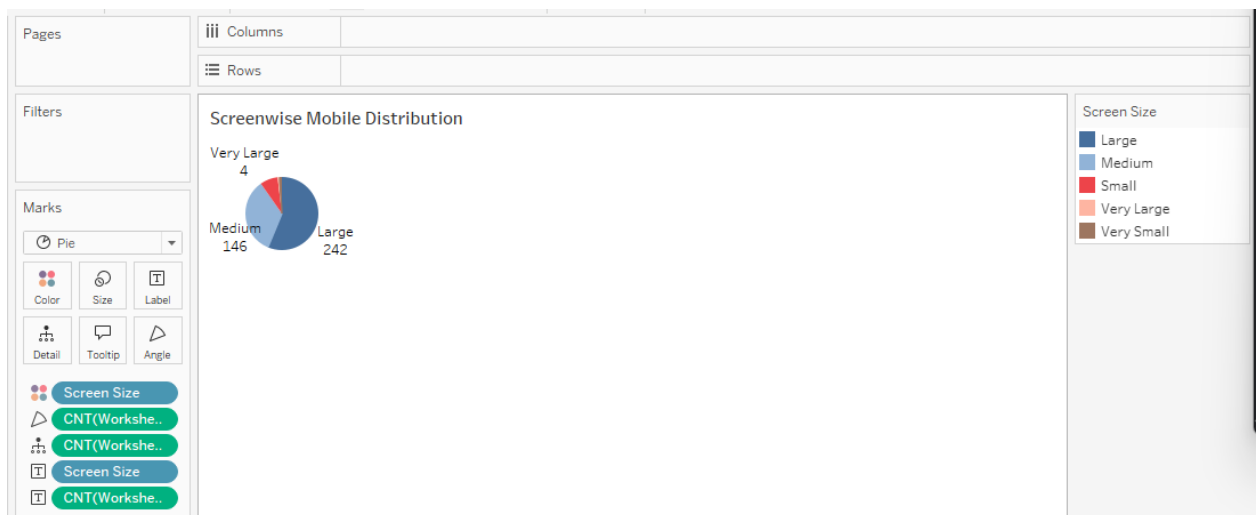
Insights: Phones with higher battery capacity (above 5000 mAh) are generally priced higher, reflecting their perceived value for heavy users.

Q. Which phone colors are most commonly available in the listings?



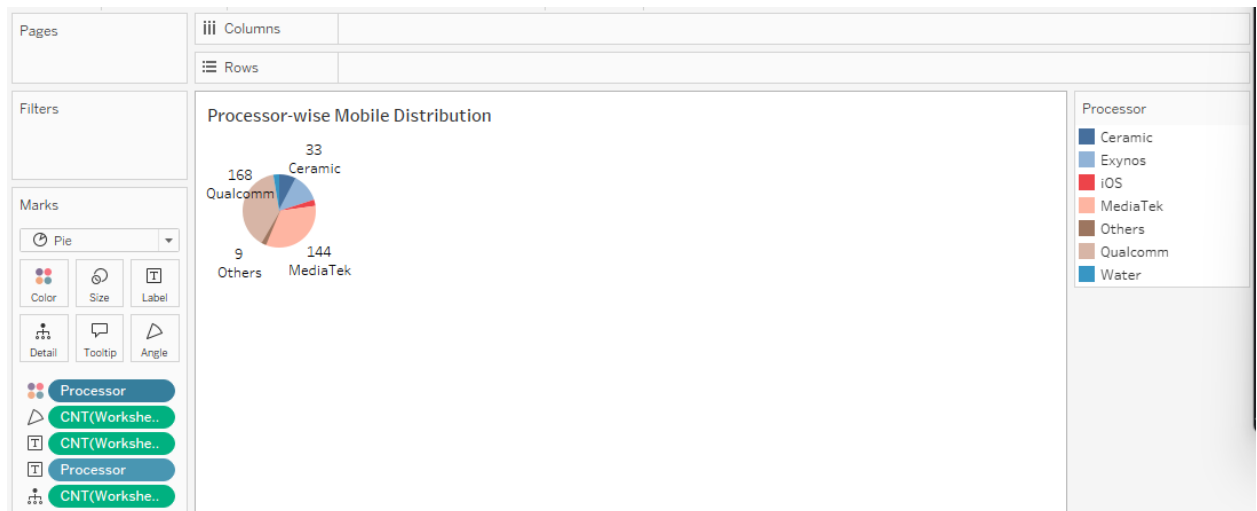
Insights: Black and Blue are the most popular base colors in listings, aligning with general consumer preferences

Q. How are mobile phones distributed based on screen size categories?



Insights: Most phones have large screen sizes between 6.3 to 6.7 inches, matching current market trends toward larger displays.

Q. Which mobile processors are most commonly used in listed smartphones?



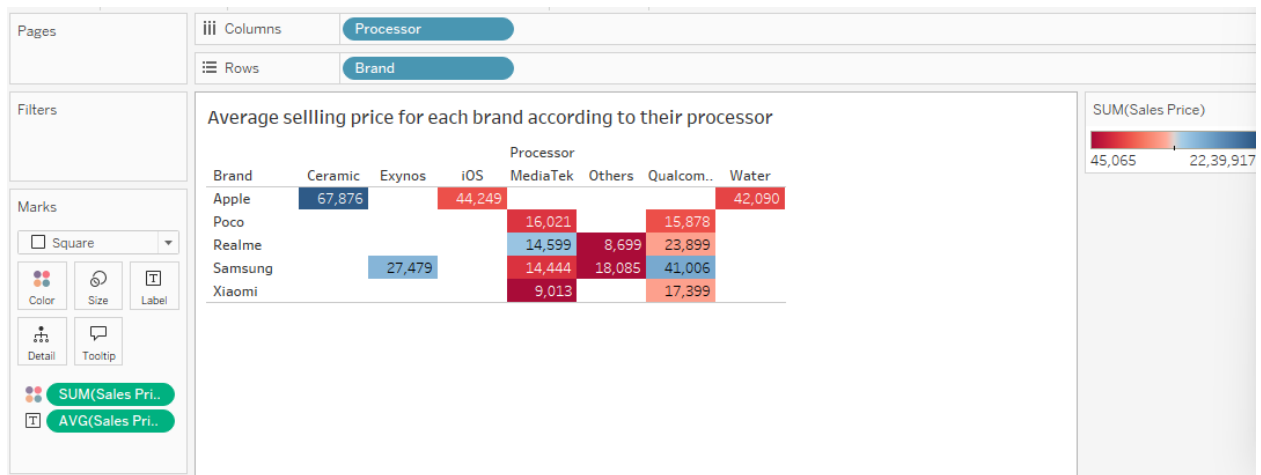
Insights: Qualcomm and MediaTek dominate the mobile processor market, with 168 and 144 models respectively, indicating their wide adoption. Ceramic and other processors have a smaller presence, suggesting niche or high-end usage.

Q. How does the number of rear cameras influence customer ratings?



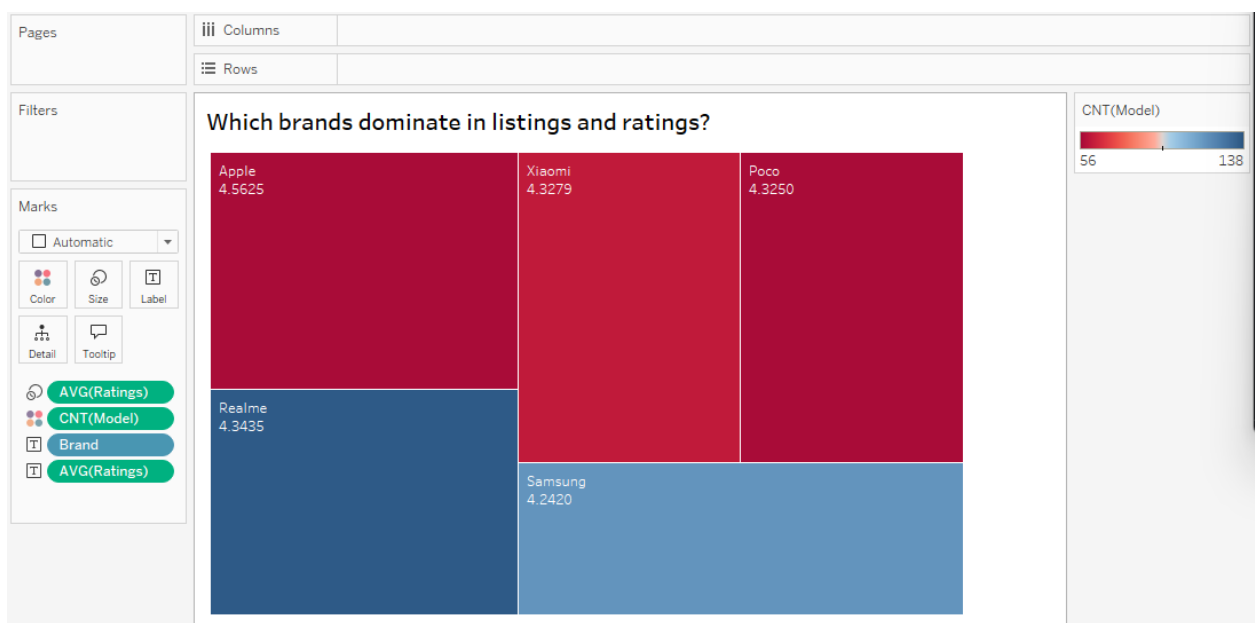
Insights: Smartphones with four rear cameras dominate the listings and consistently receive higher user ratings (4.0+), indicating strong consumer preference. Devices with fewer cameras are less common and tend to receive lower ratings, highlighting camera count as a key factor in perceived value.

Q. How does the average selling price vary by brand and processor type?



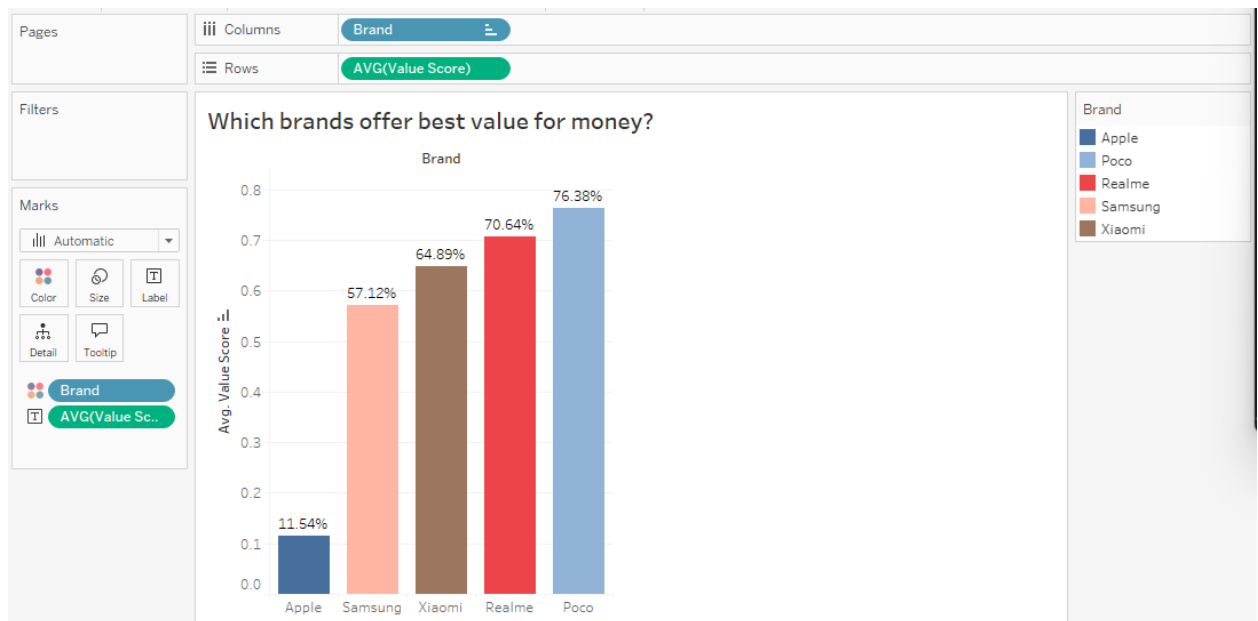
Insights: Apple’s Ceramic processor phones are the most expensive, with an average selling price of ₹67,876. Samsung’s Exynos models and Apple’s iOS line also fall into the higher price range, indicating their premium positioning. In contrast, Xiaomi and Realme maintain relatively low prices, particularly for models equipped with MediaTek and Qualcomm processors, making them attractive options for budget-conscious buyers. Poco, too, stands out by offering a balance between performance and affordability, making it a strong contender in the mid-range segment.

Q. Which brands have a strong presence in both the number of listings and high ratings?



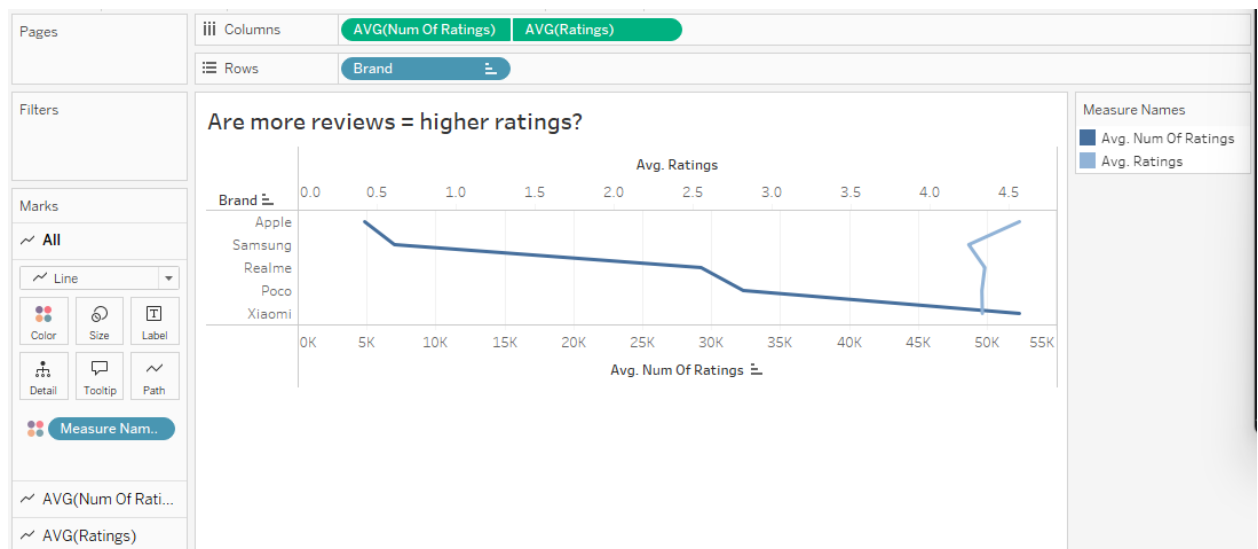
Insights: Apple leads in average rating (4.56), suggesting high customer satisfaction despite fewer models, whereas Realme and Samsung offer more model variety (larger blocks in the treemap) and Xiaomi and Poco also maintain strong ratings (~4.32–4.34), indicating overall good product reception.

Q. Which mobile brand provides the best balance between features and price?



Insights: Brands like Poco, Realme, and Xiaomi dominate the value-for-money space, making them appealing choices for budget-conscious buyers. On the other hand, Apple's low score shows that while it's a strong brand, it's not perceived as offering good value, possibly due to its high pricing strategy.

Q. Is there a positive relationship between the number of reviews and average ratings?



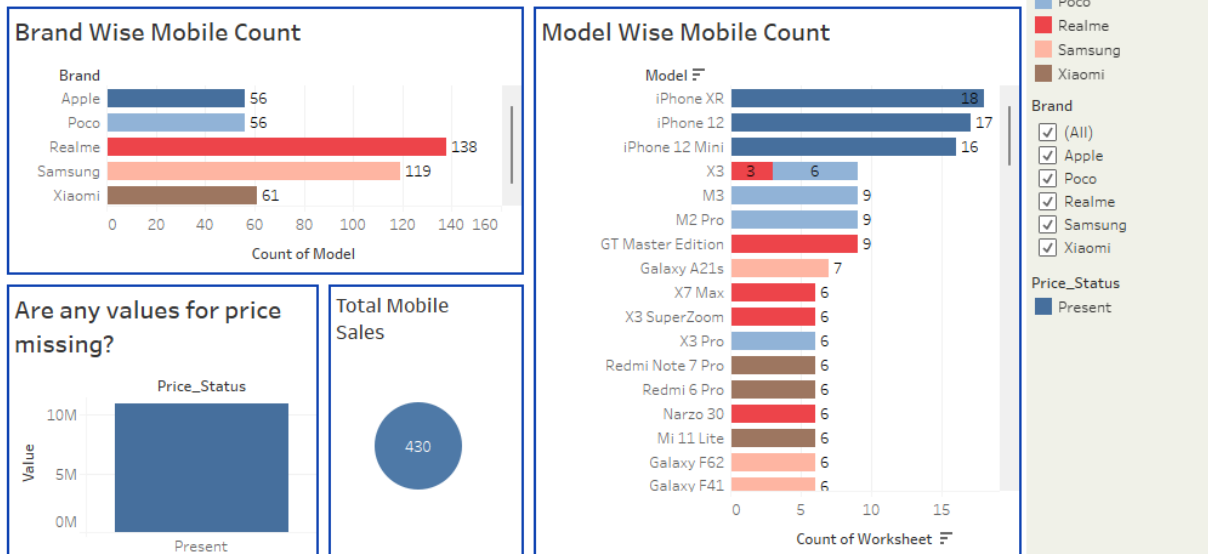
Insights: The dual-axis comparison reveals an inverse trend between popularity and satisfaction. Xiaomi, despite having the highest number of user reviews, averages lower ratings. In contrast, Apple and Poco maintain higher ratings with fewer reviews, suggesting that these brands deliver quality that resonates with a niche but satisfied user base.

Part C: Storyboard

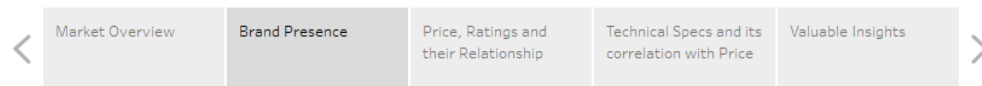
BIVA PROJECT: Flipkart Mobile Sales



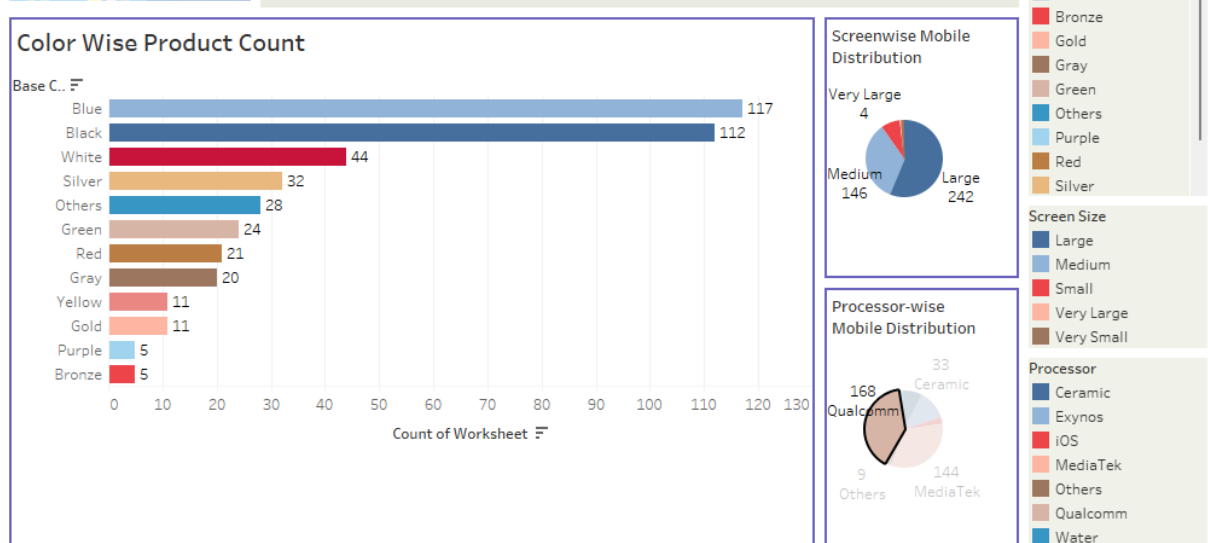
Market Overview and Brand Presence



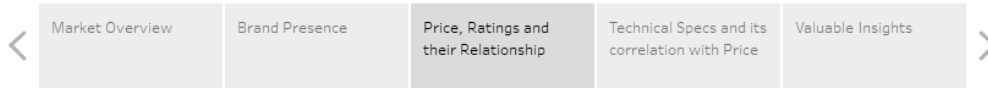
BIVA PROJECT: Flipkart Mobile Sales



Market Overview and Brand Presence

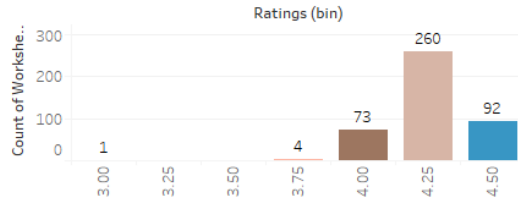


BIVA PROJECT: Flipkart Mobile Sales

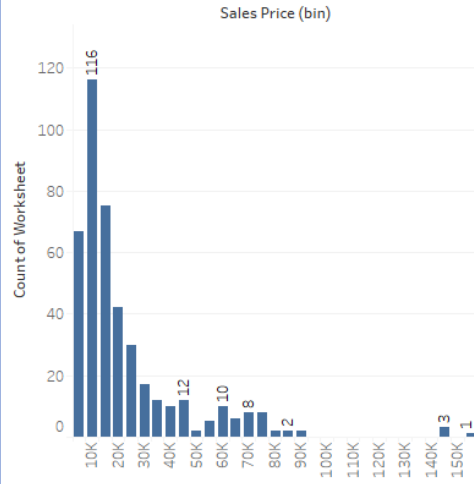


Price Ratings and Value Analysis

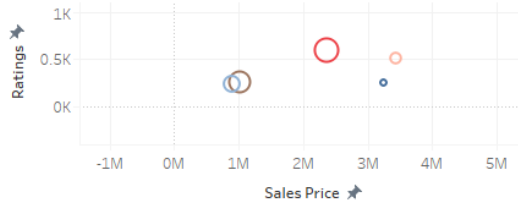
Ratings Distribution



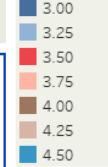
Price Distribution



Relationship between price and ratings



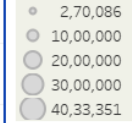
Ratings (bin)



Measure Names

Count of Worksheet

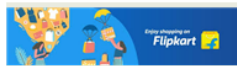
Num Of Ratings



Brand

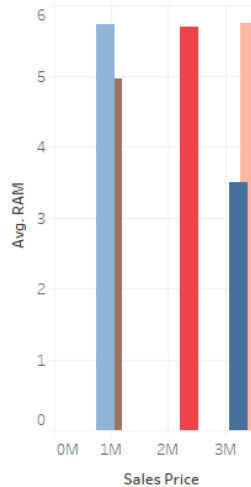


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Technical Specs v/s Price Analysis

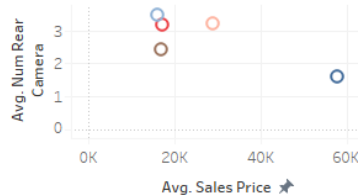
RAM v/s Price



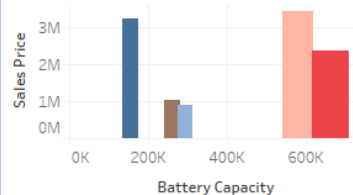
Average selling price for each brand according to their processor

Brand	Processor						
	Ceramic	Exynos	iOS	MediaTek	Others	Qualcom..	Water
Apple	67,876		44,249				42,090
Poco				16,021		15,878	
Realme				14,599	8,699	23,899	
Samsung		27,479		14,444	18,085	41,006	
Xiaomi				9,013		17,399	

Camera v/s Price v/s Brand



Battery v/s Price



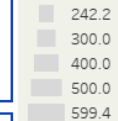
Brand



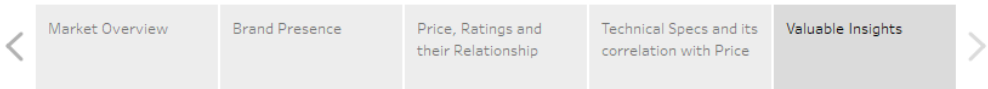
Sales Price



Ratings

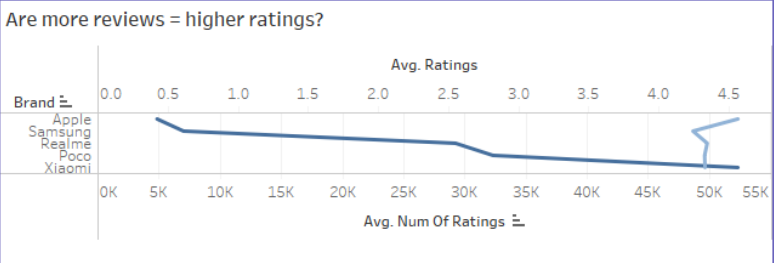
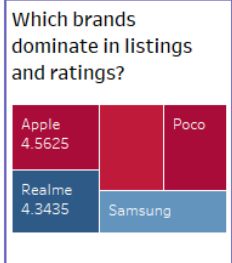


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Valuable Insights

Count of Model
56 138



Measure Names
Avg. Num Of Ratings
Avg. Ratings

Brand
Apple
Poco
Realme
Samsung
Xiaomi

